

Customer Churn Analysis



Analysis, Retention & Churn Telecom

Presented by

TEAM 3

Analyzing customer behavior to predict churn risk and enable proactive retention strategies



Team members



DATA & SQL

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MACHINE LEARNING

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PYTHON

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POWER BI

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AGENDA

- 01 Project Overview
- 02 Project Objectives
- 03 Target Audience
- 04 Challenges & Problems
- 05 Proposed Solutions
- 06 Workflow & Data Pipeline
- 07 Technical Implementation
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PROJECT OVERVIEW

Customer churn has become a major challenge for telecommunications companies as customers increasingly switch to competitors

Our project analyzes historical customer data to identify the key factors behind churn and understand customer behavior

The system then uses predictive analytics to identify customers who are at high risk of leaving



01 — ANALYZE

Analyze historical customer data to identify the root causes of churn



02 — PREDICT

Build a predictive machine learning model to calculate the churn risk score for each customer

ANALYZE → PREDICT → RETAIN

PROJECT OBJECTIVES

03 — RETAIN

Enable proactive retention through personalized recommendations and strategies before customers leave





TARGET AUDIENCE

Telecommunications companies seeking to reduce customer churn and improve customer retention

01 — RETENTION TEAMS

Identify high-risk customers and take action before they leave

02 — SALES & CUSTOMER SERVICE

Receive targeted customer lists and personalized retention recommendations

03 — BUSINESS DECISION MAKERS

Use data-driven insights to improve retention strategies and reduce churn

CHALLENGES & PROBLEMS

01 — CUSTOMER CHURN

Increasing customer churn and switching to competitors

02 — ROOT CAUSE IDENTIFICATION

Difficulty in identifying the main factors driving customers to leave

03 — HIGH-RISK CUSTOMERS

Difficulty in identifying customers who are likely to churn before they leave

04 — RETENTION ACTIONS

The need for targeted and personalized retention strategies instead of random offers



PROPOSED SOLUTIONS

01 — PREDICTIVE RISK SCORING

Use Machine Learning to calculate a churn risk score from 0–100 for each customer

02 — EARLY WARNING SYSTEM

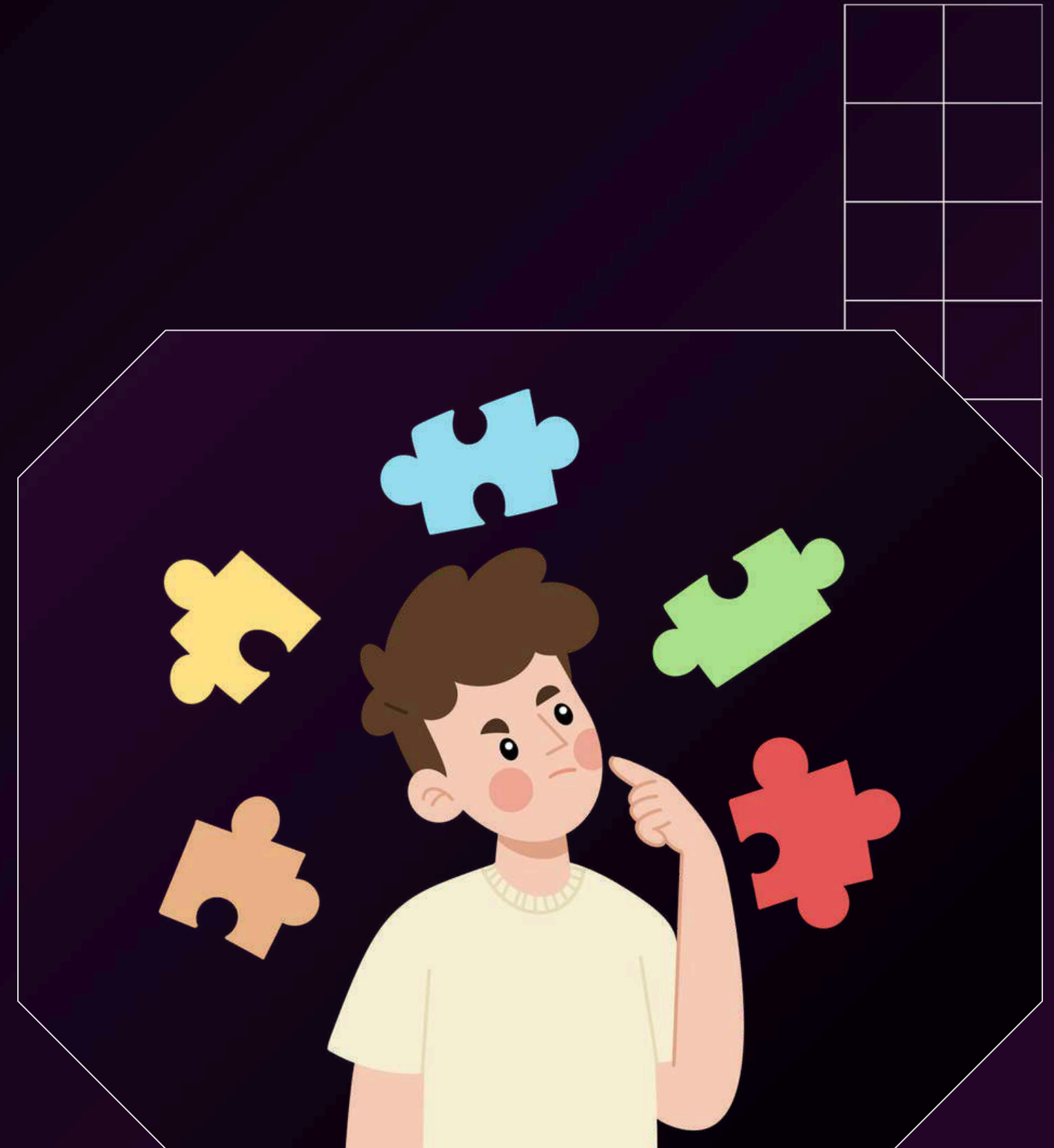
Automatically identify customers with a risk score above 70% and flag them for action

03 — PERSONALIZED OFFERS

Recommend targeted retention offers based on the reason behind each customer's risk

04 — OPERATIONAL ACTIONS

Address root causes through contract migration, prioritized technical support, and incentives for automatic payment



WORKFLOW & DATA PIPELINE

End-to-end process from raw data to actionable insights



TECHNICAL IMPLEMENTATION

SQL &
DATABASE

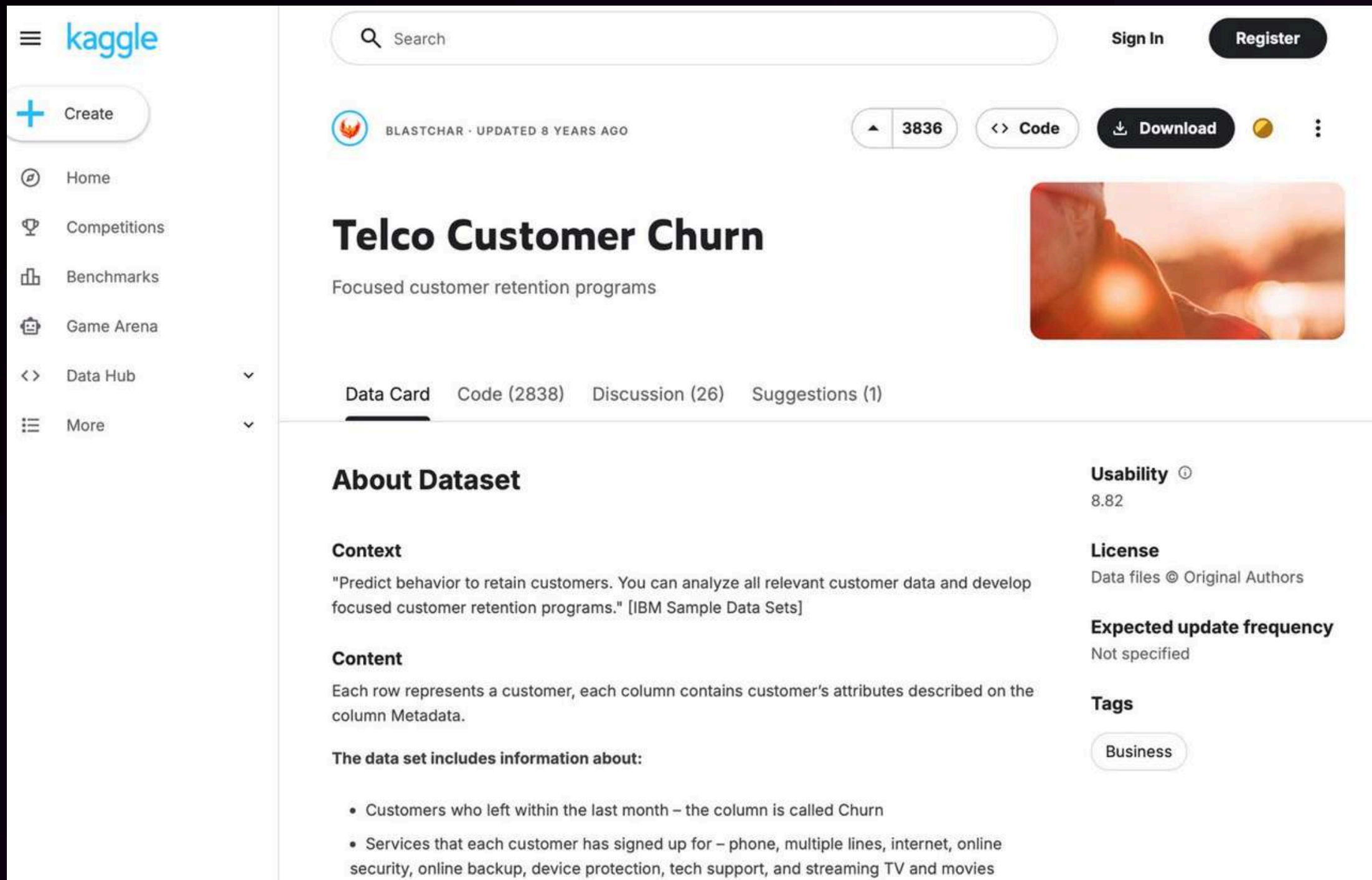
MACHINE
LEARNING

PYTHON

POWER BI &
DASHBOARD



data source



The screenshot shows the Kaggle dataset page for 'Telco Customer Churn'. The page layout includes a left sidebar with navigation links (Home, Competitions, Benchmarks, Game Arena, Data Hub, More), a top header with the Kaggle logo, a search bar, and 'Sign In'/'Register' buttons. The dataset card features the title 'Telco Customer Churn', a subtitle 'Focused customer retention programs', and a thumbnail image of a person on a phone. Below the title are tabs for 'Data Card' (selected), 'Code (2838)', 'Discussion (26)', and 'Suggestions (1)'. The 'About Dataset' section is divided into 'Context' (describing the goal of predicting customer retention), 'Content' (explaining that each row is a customer with various attributes), and 'The data set includes information about:' (listing churn status and services). On the right, a 'Usability' score of 8.82, the 'License' (Data files © Original Authors), 'Expected update frequency' (Not specified), and a 'Tags' section with a 'Business' tag are displayed.

Telco Customer Churn
Focused customer retention programs

About Dataset

Context
"Predict behavior to retain customers. You can analyze all relevant customer data and develop focused customer retention programs." [IBM Sample Data Sets]

Content
Each row represents a customer, each column contains customer's attributes described on the column Metadata.

The data set includes information about:

- Customers who left within the last month – the column is called Churn
- Services that each customer has signed up for – phone, multiple lines, internet, online security, online backup, device protection, tech support, and streaming TV and movies

Usability 8.82

License
Data files © Original Authors

Expected update frequency
Not specified

Tags
Business

Dataset :
TelcoCustomer Churn

Source:
Kaggle

Dataset:
7,043 Customers | 21 Features

Contains:

- Customer Demographics
- Services
- Contracts
- Payment & Billing
- Churn Status

SQL & Database



We used SQL Server to clean, standardize, and prepare the raw customer data for further analysis

01 — REMOVE DUPLICATES

ROW_NUMBER() + CTE

Numbered records for each customer using customerID

Used WHERE rn = 1

Kept one unique record for each c

02 — STANDARDIZE VALUES

CASE WHEN

- "No internet service" → "No"
- "No phone service" → "No"
- Standardized text values for easier analysis

03 — HANDLE DATA TYPES & MISSING VALUES

CAST + NULLIF + COALESCE

- tenure → INT
- Financial values → DECIMAL(10,2)
- Empty TotalCharges → NULL
- Missing values → 0



SQL & Database



KPI EXTRACTION & CUSTOMER ANALYSIS

After cleaning the data, we used SQL to extract key performance indicators and analyze customer behavior.

01 — RETENTION RATE

Retention Rate = $(\text{Customers with Churn} = \text{'No'} / \text{Total Customers}) \times 100$

- Measures the percentage of customers who remain with the company
- Helps evaluate customer retention performance

02 — AVERAGE METRICS

Using AVG()

- Average Customer Tenure
- Average Monthly Charges
- Average Total Charges

03 — CONTRACT DISTRIBUTION

Using GROUP BY + SUM()

OVER() grouped customers by contract type

- Calculated the percentage of each contract type
- Compared each contract category with the total



SQL & Database



DATA NORMALIZATION & DATABASE DESIGN

We transformed the original flattable into a relational model to reduce data redundancy and improve database organization and performance

OUR ENTITIES

CUSTOMERS

Personal & demographic information

Primary Key: customerID

SERVICES

Internet, phone & protection services

CONTRACTS_BILLING

Contract details, payment methods & charges

CHURN_TRACKING

Customer tenure & churn status

RELATIONSHIPS

- All related tables are connected through customerID
- Primary Key: customerID in CUSTOMERS
- Foreign Key: customerID in the related tables
- Relationship: 1 : 1



Python

DataCleaning,KeyDrivers&Financial Impact

Data Quality& Cleaning(100% Prepared):

- Analyzed 7,021 customer accounts with 100% data cleanliness.
 - Imputed missing invoice values using the median.
 - Removed duplicate entries and whitespace; verified 0 Outliers to ensure
-
- Key Churn Drivers:
 - Payment Method: Electronic Check shows the highest churn rate at 45.3%.
 - Contract Type: Month-to-Month contracts account for a 42.7% churn rate.
 - Tech Support: Absence of tech support doubles churn risk to 41.6

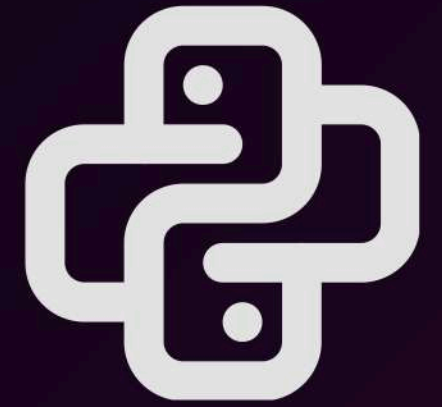
Financial Impact:

- High-Revenue Loss: Churned customers spend an average of \$74.4/month, compared to \$61.3/month for retained customers.
- Core Insight: Churn is heavily depleting the company's highest-spending customer segment

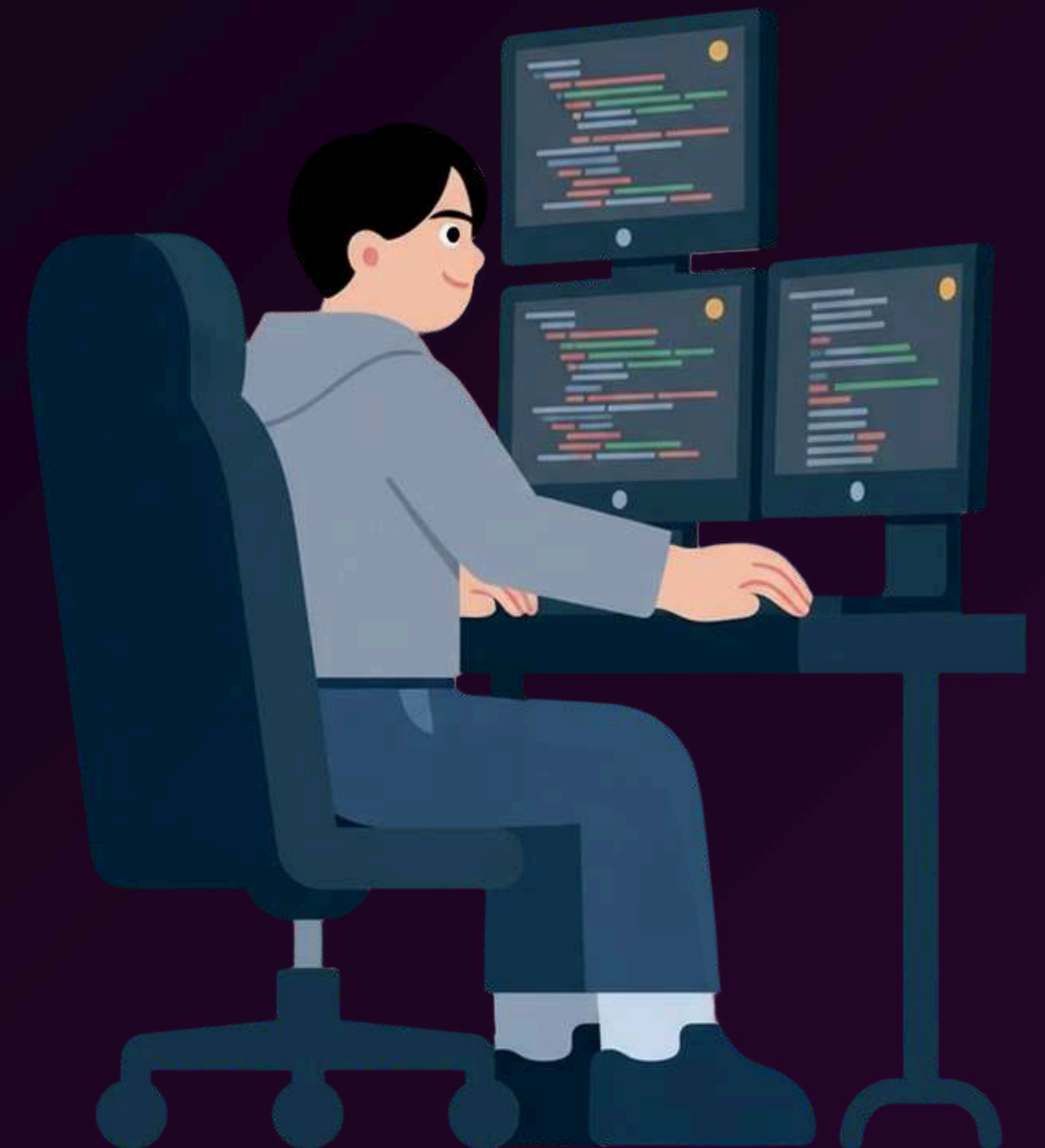


Python

Executive Recommendations & Action Plan



- Annual Contract Migration:
 - Offer targeted discounts and incentives to transition high-risk Month-to-Month users into 1-year or 2-year contracts.
- Complimentary Tech Support:
 - Provide 2 months of free technical support for new subscribers to minimize early friction and support-related churn.
- Auto-Pay Incentives:
 - Introduce bill credits to encourage users to move away from Electronic Checks toward automated payment methods (ACH / Auto-card).
- Structured Onboarding Program:
 - Activate an intensive 60-day onboarding journey for new customers to drive feature adoption and long-term satisfaction.



Machine

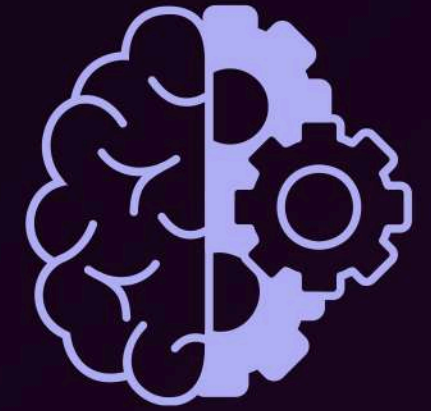
Project Overview & Data Preprocessing Pipeline

- Project Goal & Objective:
 - Build an end-to-end Machine Learning Pipeline for Customer Churn Prediction.
 - Identify at-risk customers early to proactively minimize revenue loss and improve customer retention rates.
- Data Preprocessing & Structure:
 - Type Casting & Cleaning: Converted TotalCharges to numeric data type, replaced whitespace entries with NaN, and dropped non-predictive customerID.
 - Smart Encoding Strategy: Applied LabelEncoder for binary features and One-Hot Encoding (drop_first=True) for multi-category features to prevent multicollinearity.
- Splitting & Leakage Prevention:
 - Stratified Train-Test Split: Partitioned data into 80% Train / 20% Test using stratified splitting to preserve class balance.
 - Feature Scaling: Applied StandardScaler by using fit_transform exclusively on the Training set to strictly prevent Data Leakage into the Test set.



Machine

Model Training, Evaluation & Business Impact



- Model Training & Benchmark:
- Multi-Algorithm Pipeline: Trained and evaluated 3 core classification models:
 - Logistic Regression
 - Random Forest (200 Estimators)
 - Gradient Boosting
- Evaluation Metrics: Evaluated models using Accuracy, Precision, Recall, F1-Score, and ROC-AUC while fixing `RANDOM_STATE = 42` for full reproducibility.
- Business Impact & Optimization:
 - Metric Prioritization: Focused primarily on Recall and ROC-AUC due to class imbalance and the critical cost of missing actual churners.
 - Minimizing False Negatives: Leveraged Confusion Matrix analysis to reduce False Negatives, directly protecting high-value account revenue and optimizing targeted retention campaigns.



Power bi



Dashboard Overview & Analytics Lifecycle

- **Project Scope & Objective:**
- Analyzed 7,043 customer records to identify churn patterns and customer retention opportunities.
- **Data Cleaning & ETL (Power Query):**
- Handled missing values, standardized data formats, and enforced accurate data types across all fields
- **Data Modeling & Key Metrics:**
- Created a relational schema using unique customerID keys.
- Engineered core KPIs and interactive dashboard visuals for real-time tracking.



Conclusion

- Our project analyzes customer behavior, identifies the key causes of churn, and predicts customers at high risk of leaving
- By combining SQL, Python, Machine Learning, and Power BI, we turn customer data into actionable retention strategies

ANALYZE • PREDICT • RETAIN

SQL • Python • Machine Learning • PowerBI

Our goal was not only to understand customer churn, but to predict it early and turn that prediction into action





TELCO ANALYTICS



ERD



SQL



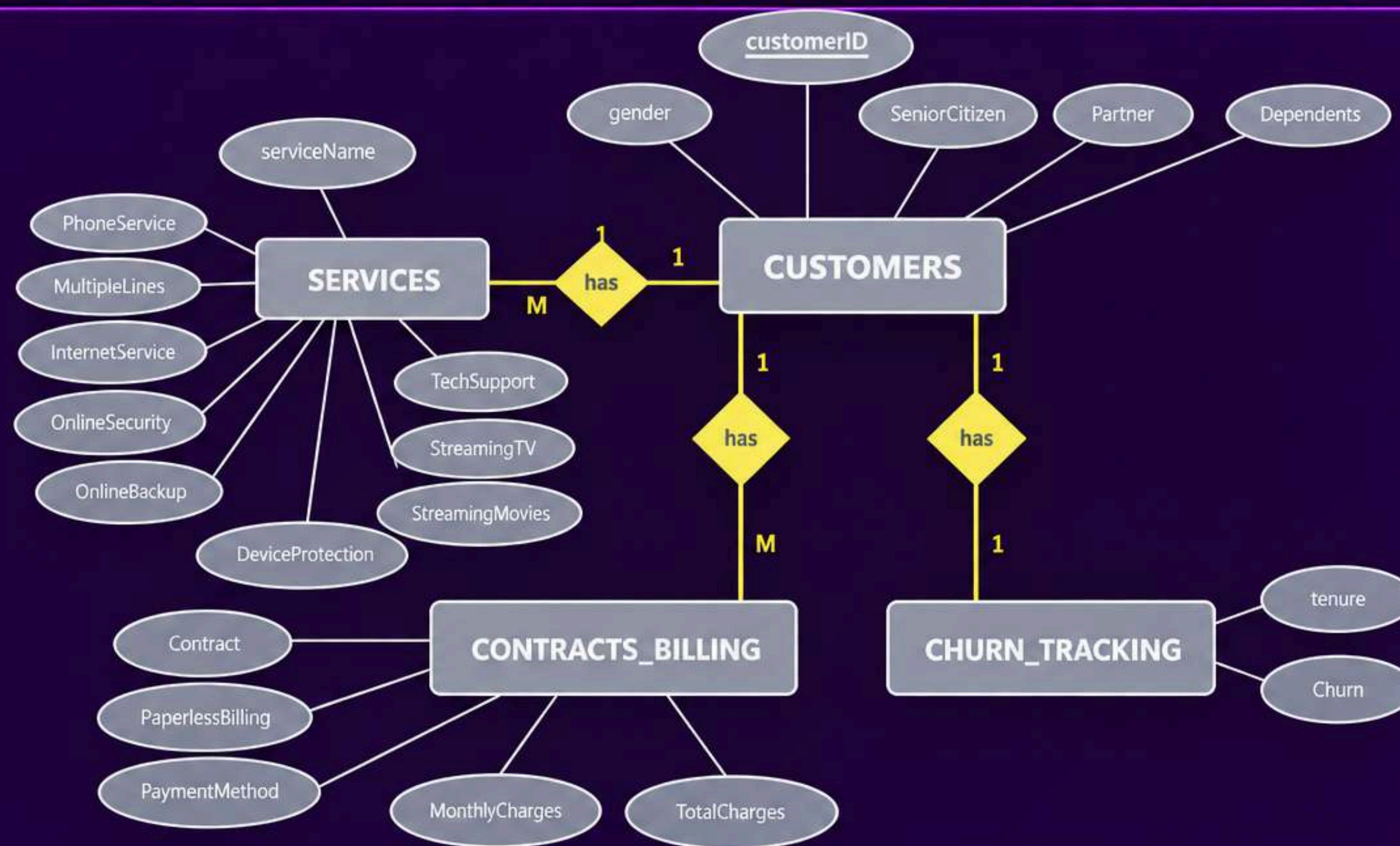
Python



AI Model



Dashboard



Database Design & Architecture Summary

- **Schema Setup:** Created TelecomDB in SQL Server, divided into four core tables: CUSTOMERS, SERVICES, CONTRACTS_BILLING, and CHURN_TRACKING.
- **Core Relationships:** CUSTOMERS serves as the primary entity (customerID PK), linked via Foreign Keys to SERVICES (1:M), CONTRACTS_BILLING (1:M), and CHURN_TRACKING (1:1).
- **Data Pipeline:** Executed schema creation via DDL and populated tables directly from clean_telecom_data using INSERT INTO ... SELECT.



TELCO ANALYTICS



ERD



SQL



Payton



AI Model



Dashboard



CHURN_TRACKING

churnID
customerID
tenure
Churn

CHURN_TRACKING

CUSTOMERS

customerID
gender
SeniorCitizen
Partner
Dependents

SERVICES

serviceID
customerID
PhoneService
MultipleLines
InternetService
OnlineSecurity
OnlineBackup
DeviceProtection
TechSupport
StreamingTV
StreamingMovies

CONTRACTS BILLING

billingID
customerID
Contract
PaperlessBilling
PaymentMethod
MonthlyCharges
TotalCharges

Technical Overview & Business Impact



- **Data Cleaning & Preprocessing:** Handled missing values, standardized text formats, and eliminated duplicate records using Common Table Expressions (CTEs) and ROW_NUMBER().



- **KPI & Feature Analytics:** Extracted key retention metrics, average billing patterns, and contract distribution across segments using SQL Window Functions.



- **Database Normalization (ERD):** Restructured the flat dataset into a normalized 4-table relational schema to optimize query performance and eliminate data redundancy.



- **Business Impact:** Bridged technical data engineering with business analytics to provide actionable insights that empower management to mitigate customer churn.

Access the complete code and
documentation here:



GITHUB



TELCO ANALYTICS



ERD



SQL



Python



AI Model



Dashboard



7.043K

Total customers



\$64.76

MonthlyCharges



32.37

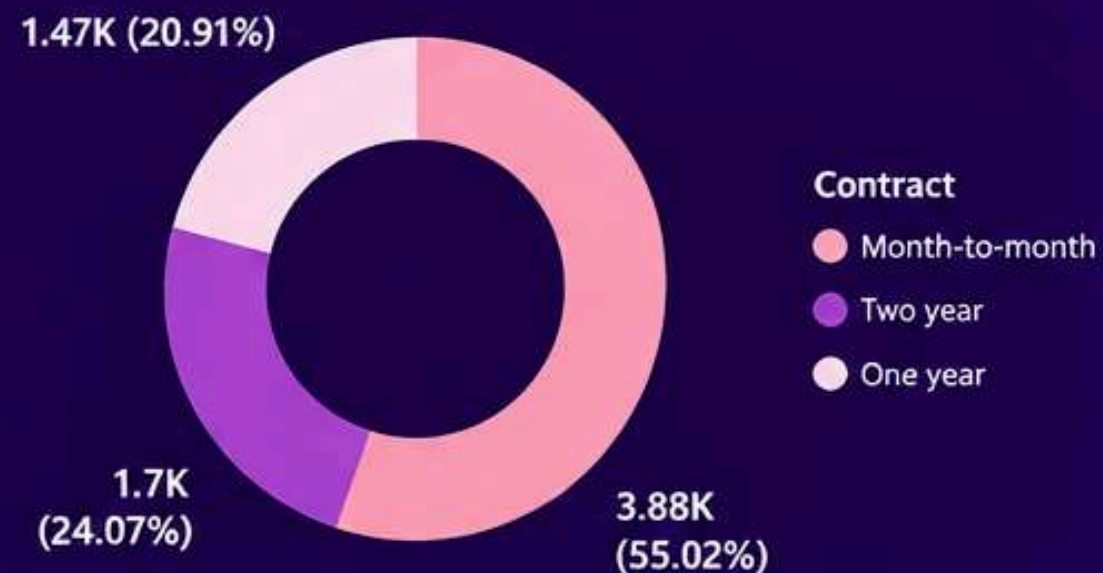
Avg of tenure



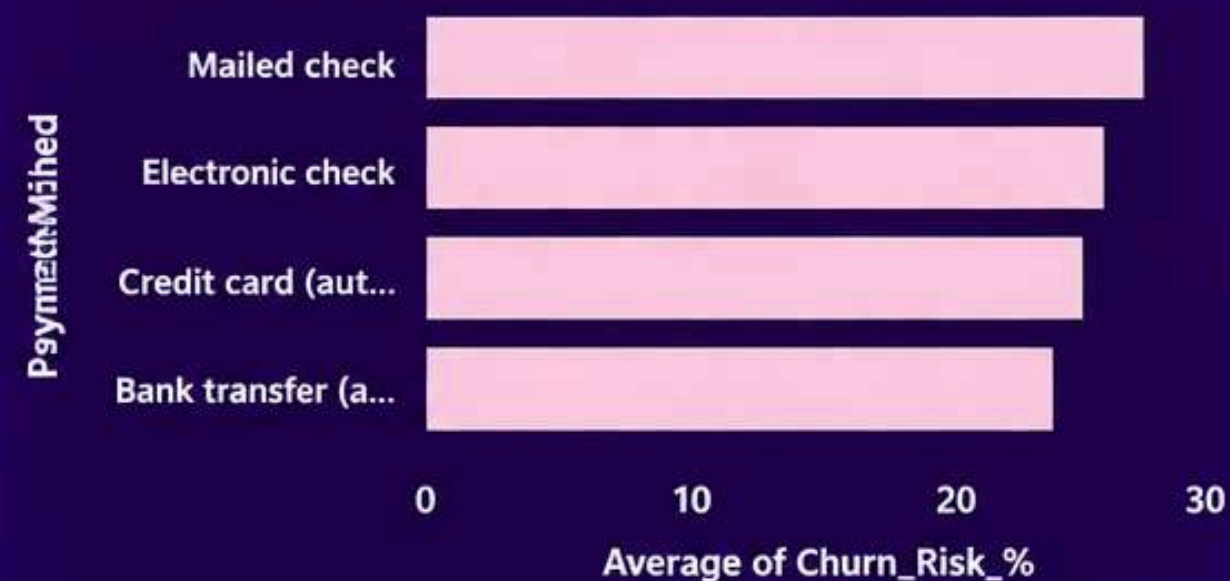
26.78

Churn_Risk_%

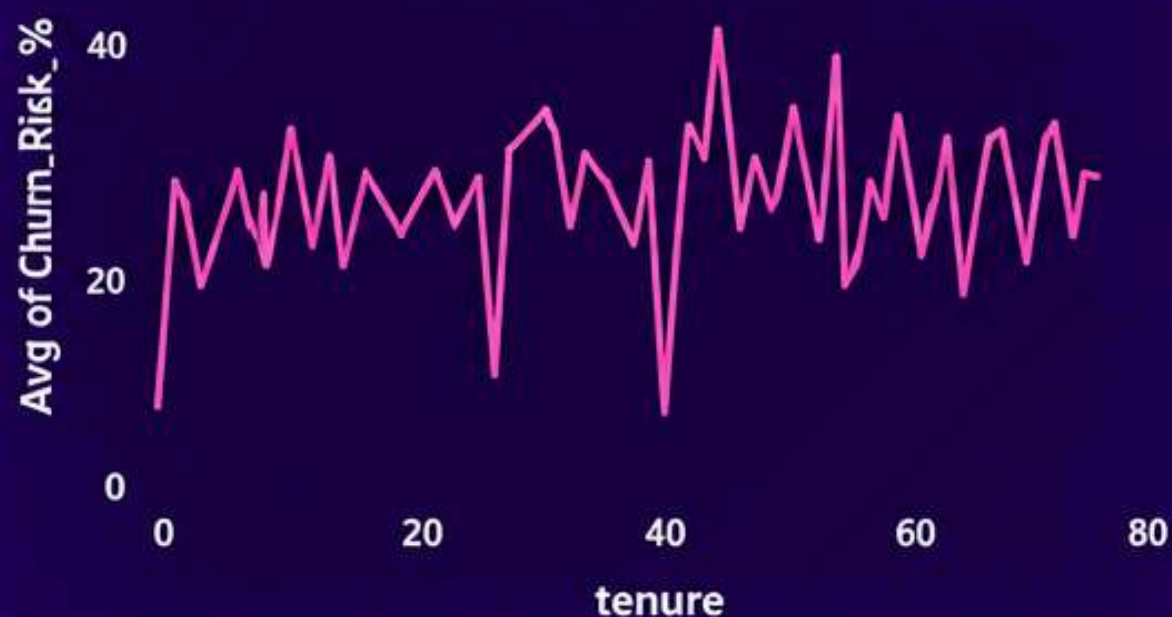
Customers By Contract Type



Churn_Risk by PaymentMethod



Avg of Churn_Risk_% by tenure

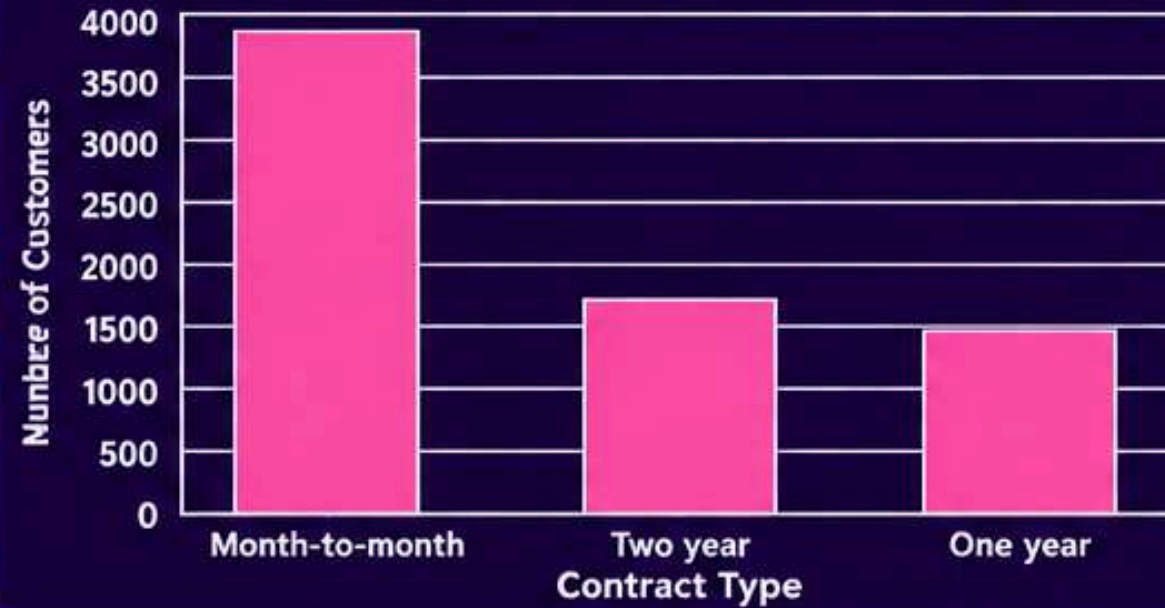


Count of customerID by OnlineSecurity and Churn

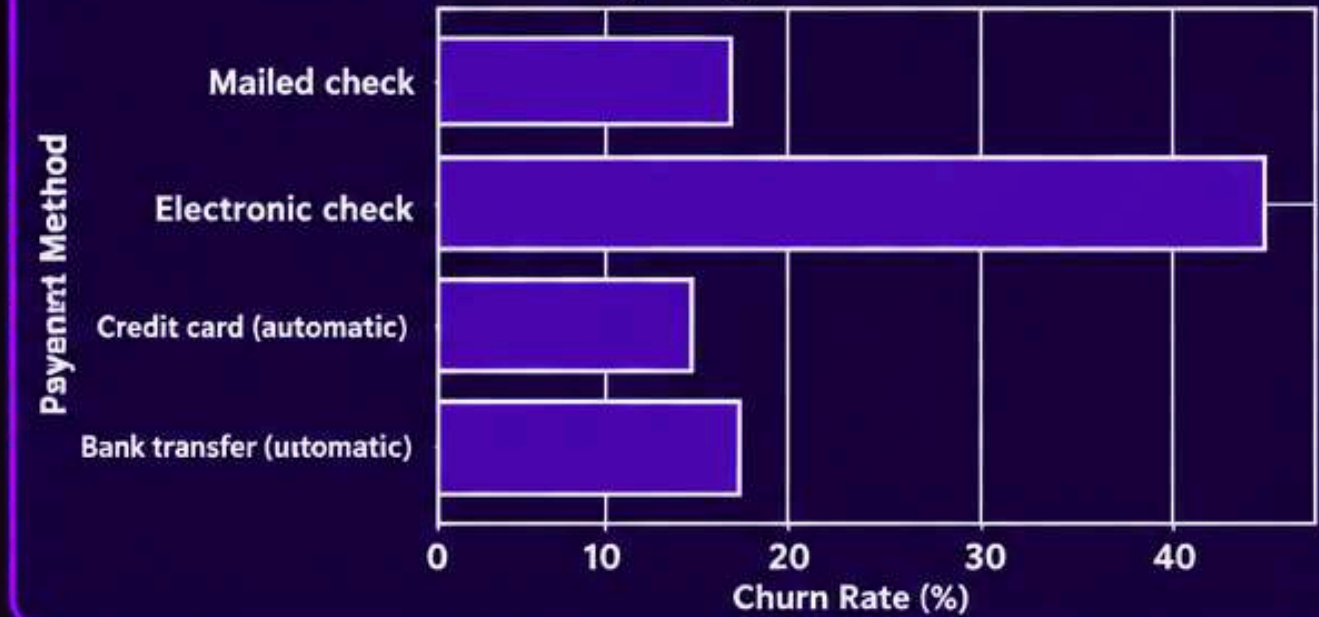




Number of Customers per Contract Type



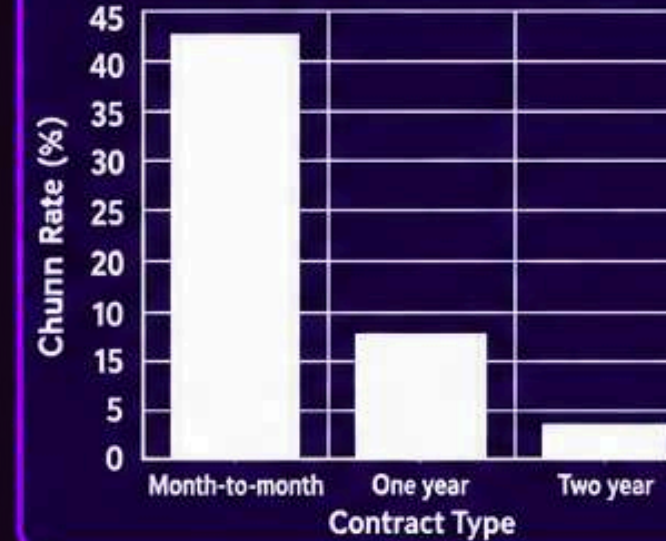
Churn Rate by Payment Method



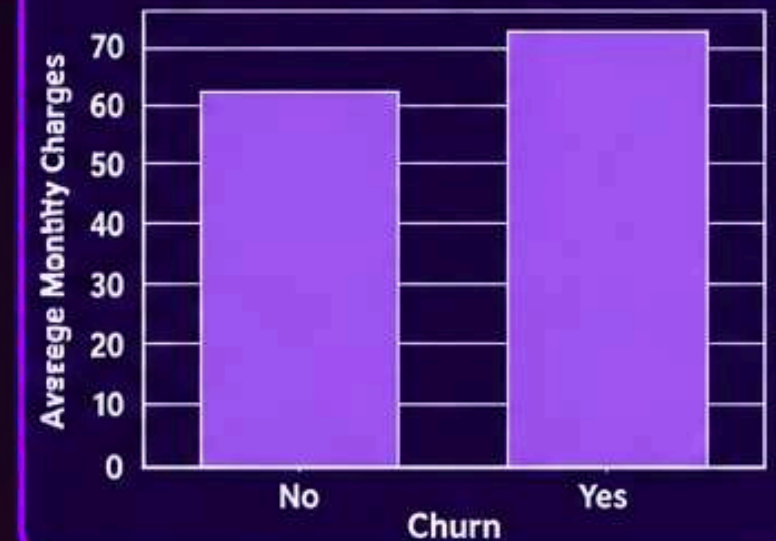
Monthly Charges Distribution by Payment Method



Churn Rate by Contract Type



Average Monthly Charges by Churn Status



Key Analytical Focus Areas

- **Contract Types:** Impact of short-term vs. long-term commitments on churn.
- **Support Services:** Effect of technical support availability on retention.
- **Billing & Payments:** Relationship between monthly charges, payment options, and customer loss.

Business Value

- Early identification of at-risk customers.
- Data-driven retention and loyalty strategies.

Access the complete code and documentation here:





TELCO ANALYTICS



ERD



SQL



Python



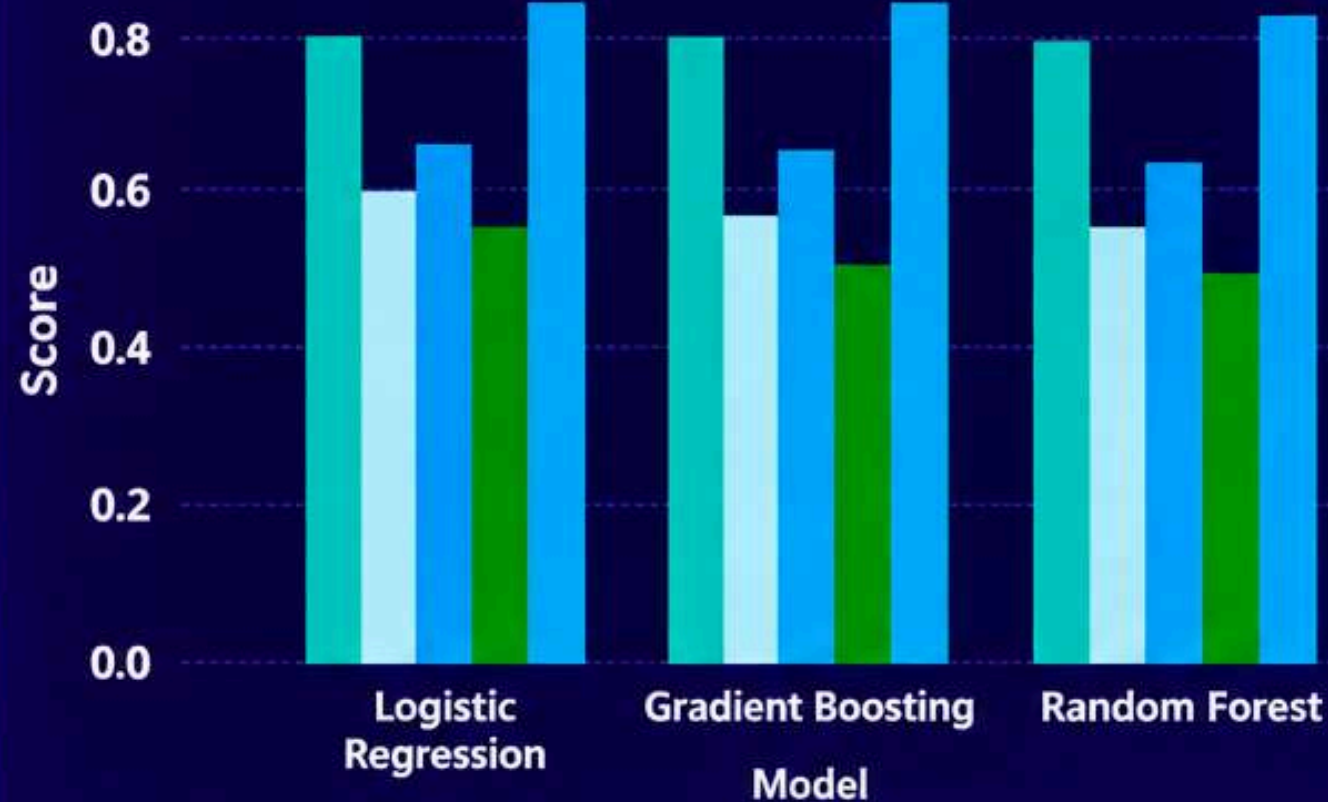
AI Model



Dashboard



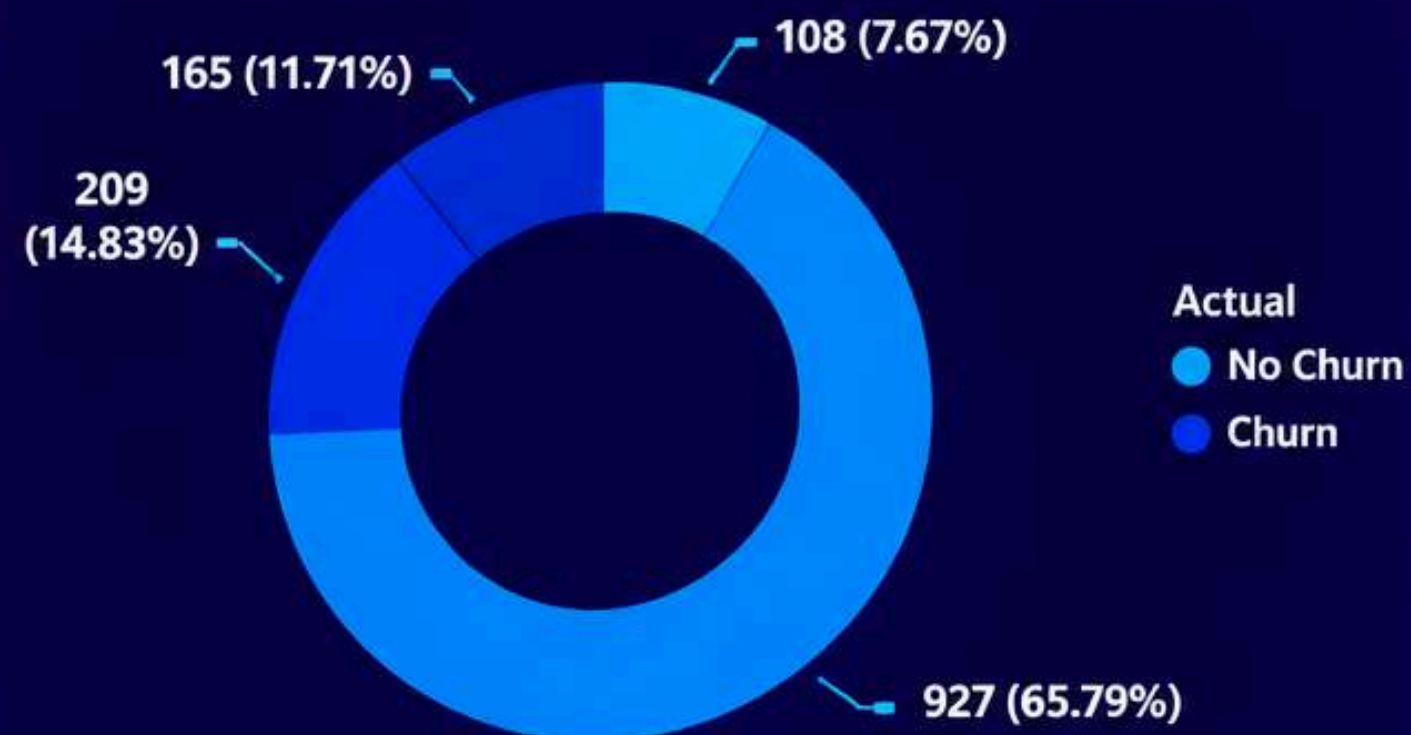
Sum of Accuracy Sum of F1-Score Sum of Precision Sum of Recall Sum of ROC-AUC



Model Performance Summary

- Top Model:** Logistic Regression achieves the highest performance with **80.6% Accuracy**.
- Key Metrics:** Leads across Precision and F1-Score compared to Gradient Boosting and Random Forest.
- Recommendation:** Best suited model for deployment to predict customer churn reliably.

Sum of Value by Actual and Predicted



Logistic Regression Matrix Insights

- High Accuracy:** Correctly identified **927 Retained** and **209 Churned** customers.
- Low Error Rate:** Only **165 false negatives** (churn risks missed by the model).

Access the complete code and documentation here:



THE SPECIAL THANKS & GUIDANCE

INSTRUCTOR : ENG/GHARAM MOKHTAR